

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

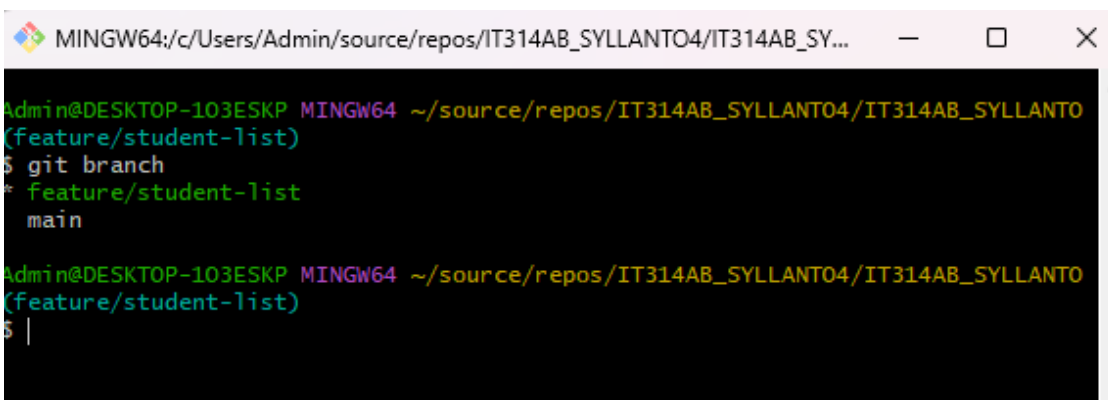
Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch



```
MINGW64:/c:/Users/Admin/source/repos/IT314AB_SYLLANTO4/IT314AB_SY...  
Admin@DESKTOP-103ESKP MINGW64 ~/source/repos/IT314AB_SYLLANTO4/IT314AB_SYLLANTO  
(feature/student-list)  
$ git branch  
* feature/student-list  
main  
Admin@DESKTOP-103ESKP MINGW64 ~/source/repos/IT314AB_SYLLANTO4/IT314AB_SYLLANTO  
(feature/student-list)  
$ |
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```
19
20 class Student {
21   final String image;
22   final String name;
23   final String course;
24   final String yearLevel;
25   final int age;
26   final String hobby;
27
28   const Student({
29     required this.image,
30     required this.name,
31     required this.course,
32     required this.yearLevel,
33     required this.age,
34     required this.hobby,
35   });
36 }
37
38 const List<Student> students = [
39   Student(
40     image: 'assets/student1.jpg',
41     name: 'Mary Jocelyn Syllanto',
42     course: 'BSIT',
43     yearLevel: '3rd Year',
44     age: 20,
45     hobby: 'Reading',
46   ),
47   Student(
48     image: 'assets/student2.jpg',
49     name: 'Aaron Cruz',
50     course: 'BSCS',
51     yearLevel: '2nd Year'
```

```
48     image: 'assets/student2.jpg',
49     name: 'Aaron Cruz',
50     course: 'BSCS',
51     yearLevel: '2nd Year',
52     age: 19,
53     hobby: 'Gaming',
54   ),
55   Student(
56     image: 'assets/student3.jpg',
57     name: 'John Reyes',
58     course: 'BSIT',
59     yearLevel: '4th Year',
60     age: 21,
61     hobby: 'Basketball',
62   ),
63   Student(
64     image: 'assets/student4.jpg',
65     name: 'Sophia Lim',
66     course: 'BSIS',
67     yearLevel: '1st Year',
68     age: 18,
69     hobby: 'Painting',
70   ),
71   Student(
72     image: 'assets/student5.jpg',
73     name: 'Maria Santos',
74     course: 'BSIT',
75     yearLevel: '3rd Year',
76     age: 20,
77     hobby: 'Singing',
78   ),
79 ];
```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)

```
Card(  
  color: Colors.white.withOpacity(0.3),  
  elevation: 0,  
  child: Padding(  
    padding: const EdgeInsets.all(8),  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.center,  
      children: [  
        const SizedBox(height: 50),  
        Image.asset('assets/dog.webp', width: 130),  
        const SizedBox(height: 30),  
        Text(  
          'Mary Jocelyn Syllanto',  
          style: const TextStyle(  
            fontSize: 20,  
            fontWeight: FontWeight.bold,  
          ), // TextStyle  
        ), // Text // Text  
        const SizedBox(height: 10),  
        Text('BSIT', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 10),  
        Text('3rd Year', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 10),  
        Text('Age: 20', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 10),  
        Text('Hobby: Drinking Secret', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 50),  
      ],  
    ), // Column // Column  
  ), // Padding // Padding  
), // Card // Card
```

```
Card(  
  color: Colors.white.withOpacity(0.3),  
  elevation: 0,  
  child: Padding(  
    padding: const EdgeInsets.all(8),  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.center,  
      children: [  
        const SizedBox(height: 50),  
        Image.asset('assets/marianne.JPG', width: 130),  
        const SizedBox(height: 30),  
        Text(  
          'Maryan Gridwan',  
          style: const TextStyle(  
            fontSize: 20,  
            fontWeight: FontWeight.bold,  
          ), // TextStyle  
        ), // Text // Text  
        const SizedBox(height: 10),  
        Text('BSIT', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 10),  
        Text('3rd Year', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 10),  
        Text('Age: 21', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 10),  
        Text('Hobby: Playing Mobile Legends', style: const TextStyle(fontSize: 18)),  
        const SizedBox(height: 50),  
      ],  
    ), // Column // Column  
  ), // Padding // Padding  
), // Card // Card
```

```

Card(
  color: Colors.white.withOpacity(0.3),
  elevation: 0,
  child: Padding(
    padding: const EdgeInsets.all(8),
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        const SizedBox(height: 50),

        Image.asset('assets/mav1.JPG', width: 130),

        const SizedBox(height: 30),

        Text(
          'Mav Sumalangitka',
          style: const TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ), // TextStyle
        ), // Text // Text

        const SizedBox(height: 10),

        Text('BSCS', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

        Text('2nd Year', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

        Text('Age: 19', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),
        Text('Hobby: Dancing', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 50),
      ],
    ), // Column // Column
  ), // Padding // Padding
), // Card // Card

```

```

Card(
  color: Colors.white.withOpacity(0.3),
  elevation: 0,
  child: Padding(
    padding: const EdgeInsets.all(8),
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        const SizedBox(height: 50),

        Image.asset('assets/chim.PNG', width: 130),

        const SizedBox(height: 30),

        Text(
          'Chim Boop',
          style: const TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ), // TextStyle
        ), // Text // Text

        const SizedBox(height: 10),

        Text('BSIT', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

        Text('2nd Year', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

        Text('Age: 20', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),
        Text('Hobby: Hobby Tanan', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 50),
      ],
    ), // Column // Column
  ), // Padding // Padding
), // Card // Card

```

```

Card(
  color: Colors.white.withOpacity(0.3),
  elevation: 0,
  child: Padding(
    padding: const EdgeInsets.all(8),
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        const SizedBox(height: 50),

        Image.asset('assets/krylle.PNG', width: 130),

        const SizedBox(height: 30),

        Text(
          'Karylle Buta',
          style: const TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ), // TextStyle
        ), // Text // Text

        const SizedBox(height: 10),

        Text('BSIS', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

        Text('3rd Year', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

        Text('Age: 21', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 10),

```

```

        Text('Hobby: Magpakatanga', style: const TextStyle(fontSize: 18)),

        const SizedBox(height: 50),
      ],
    ),

```

🚩 Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's **ListView**.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

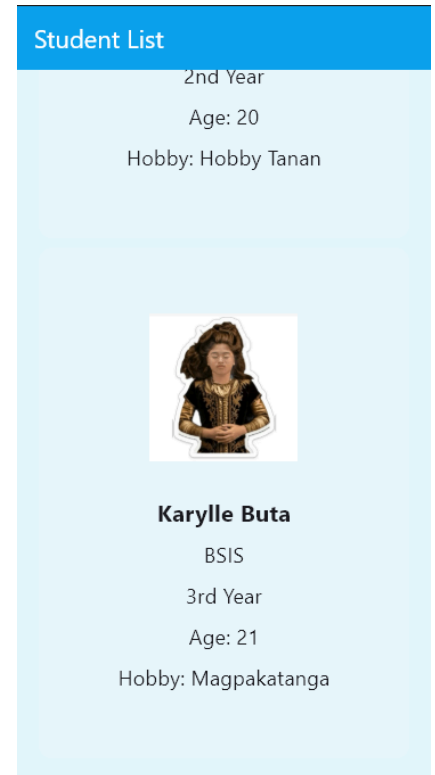
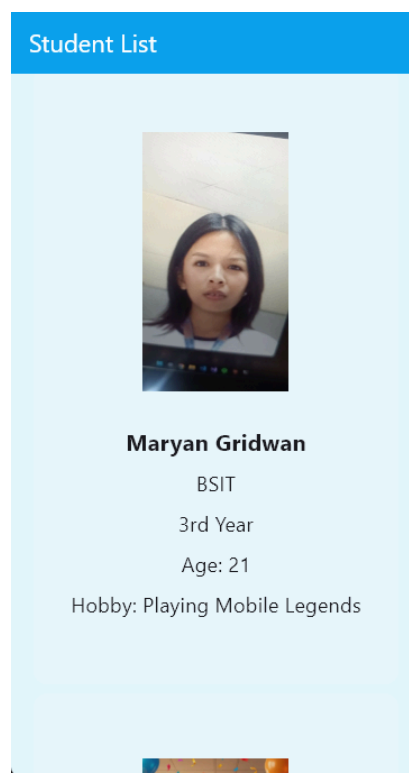
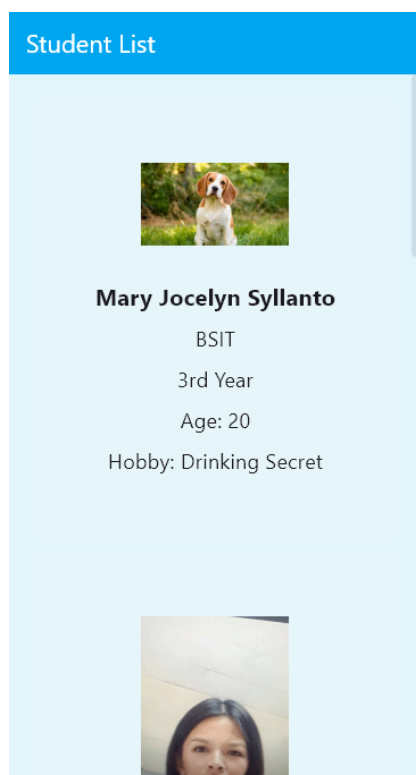
The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- **ListView** in your code
- Top of the application
- Bottom portion showing later students

```
95 | body: ListView(  
96 |   padding: const EdgeInsets.all(16),  
97 |   children: [
```



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

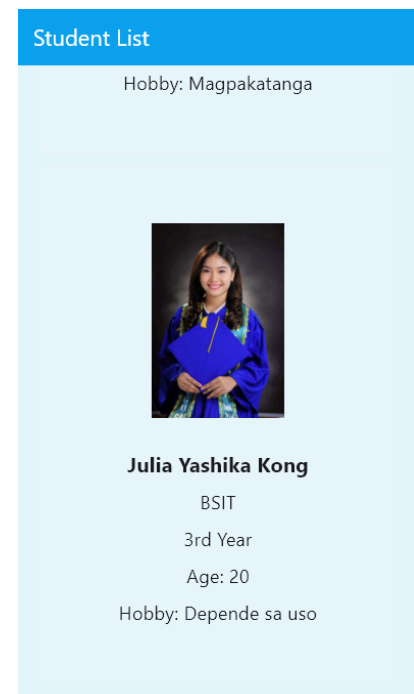
Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
103 body: ListView.builder(  
104   padding: const EdgeInsets.all(16),  
105   itemCount: students.length,  
106   itemBuilder: (context, index) {  
107     final student = students[index];
```

```
37  
38 const List<Student> students = [  
39   Student(  
40     image: 'assets/dog.webp',  
41     name: 'Mary Jocelyn Syllanto',  
42     course: 'BSIT',  
43     yearLevel: '3rd Year',  
44     age: 20,  
45     hobby: 'Drinking Secret',  
46   ),  
47   Student(  
48     image: 'assets/marianne.JPG',  
49     name: 'Maryan Gridwan',  
50     course: 'BSIT',  
51     yearLevel: '3rd Year',  
52     age: 21,  
53     hobby: 'Playing Mobile Legends',  
54   ),  
55   Student(  
56     image: 'assets/mav1.JPG',  
57     name: 'Mav Sumalangiitka',  
58     course: 'BSCS',  
59     yearLevel: '1st Year',  
60     age: 19,  
61     hobby: 'Dancing',  
62   ),  
63   Student(  
64     image: 'assets/chim.PNG',  
65     name: 'Chim Boop',  
66     course: 'BSIT',  
67     yearLevel: '2nd Year',  
68     age: 20,  
69     hobby: 'Hobby Tanan',  
70   ),
```

```
71 Student(  
72   image: 'assets/krylle.PNG',  
73   name: 'Karylle Buta',  
74   course: 'BSIS',  
75   yearLevel: '3rd Year',  
76   age: 21,  
77   hobby: 'Magpakatanga',  
78 ),  
79 Student(  
80   image: 'assets/jolya.JPG',  
81   name: 'Julia Yashika Kong',  
82   course: 'BSIT',  
83   yearLevel: '3rd Year',  
84   age: 20,  
85   hobby: 'Depende sa uso',  
86 ),  
87 ];
```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
20 class Student {
21     final String image;
22     final String name;
23     final String course;
24     final String yearLevel;
25     final int age;
26     final String hobby;
27     final String studentId;
28     final String email;
29     final String favoriteSubject;
30
31     const Student({
32         required this.image,
33         required this.name,
34         required this.course,
35         required this.yearLevel,
36         required this.age,
37         required this.hobby,
38         required this.studentId,
39         required this.email,
40         required this.favoriteSubject,
41     });
42 }
```

```
44 const List<Student> students = [
45     Student(
46         image: 'assets/dog.webp',
47         name: 'Mary Jocelyn Syllanto',
48         course: 'BSIT',
49         yearLevel: '3rd Year',
50         age: 20,
51         hobby: 'Drinking Secret',
52         studentId: '202300123',
53         email: 'mary.syllanto@school.edu',
54         favoriteSubject: 'Math',
55     ),
56     Student(
57         image: 'assets/marianne.JPG',
58         name: 'Maryan Gridwan',
59         course: 'BSIT',
60         yearLevel: '3rd Year',
61         age: 21,
62         hobby: 'Playing Mobile Legends',
63         studentId: '202300124',
64         email: 'maryan.gridwan@school.edu',
65         favoriteSubject: 'Networking',
66     ),
67     Student(
68         image: 'assets/mav1.JPG',
69         name: 'Mav Sumalangitka',
70         course: 'BSCS',
71         yearLevel: '1st Year',
```

```
72         age: 19,
73         hobby: 'Dancing',
74         studentId: '202300125',
75         email: 'mav.sumalangitka@school.edu',
76         favoriteSubject: 'Recess',
77     ),
78     Student(
79         image: 'assets/chim.PNG',
80         name: 'Chim Boop',
81         course: 'BSIT',
82         yearLevel: '2nd Year',
83         age: 20,
84         hobby: 'Hobby Tanan',
85         studentId: '202300126',
86         email: 'chim.boop@school.edu',
87         favoriteSubject: 'Networking',
88     ),
89     Student(
90         image: 'assets/krylle.PNG',
91         name: 'Karylle Buta',
92         course: 'BSIS',
93         yearLevel: '3rd Year',
94         age: 21,
95         hobby: 'Magpakatanga',
96         studentId: '202300127',
97         email: 'karylle.but@school.edu',
98         favoriteSubject: 'Information Systems',
99     ),
```

```
100     Student(
101         image: 'assets/jolya.JPG',
102         name: 'Julia Yashika Kong',
103         course: 'BSIT',
104         yearLevel: '3rd Year',
105         age: 20,
106         hobby: 'Depende sa uso',
107         studentId: '202300128',
108         email: 'julia.kong@school.edu',
109         favoriteSubject: 'GEC',
110     ),
111 ];
```

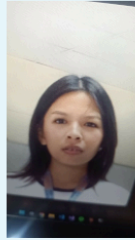
Student List



Mary Jocelyn Syllanto

BSIT
3rd Year
Age: 20
Hobby: Drinking Secret
ID: 202300123
mary.syllanto@school.edu
Favorite Subject: Math

Student List



Maryan Gridwan

BSIT
3rd Year
Age: 21
Hobby: Playing Mobile Legends
ID: 202300124
maryan.gridwan@school.edu
Favorite Subject: Networking

Student List



Mav Sumalangitka

BSCS
1st Year
Age: 19
Hobby: Dancing
ID: 202300125
mav.sumalangitka@school.edu
Favorite Subject: Recess

Student List



Chim Boop

BSIT
2nd Year
Age: 20
Hobby: Hobby Tanan
ID: 202300126
chim.boop@school.edu
Favorite Subject: Networking

Student List



Karylle Buta

BSIS
3rd Year
Age: 21
Hobby: Magpakatanga
ID: 202300127
karylle.but@school.edu
Favorite Subject: Information Systems

Student List



Julia Yashika Kong

BSIT
3rd Year
Age: 20
Hobby: Depende sa uso
ID: 202300128
julia.kong@school.edu
Favorite Subject: GEC

🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

- "No students found."
- "Student list is empty."

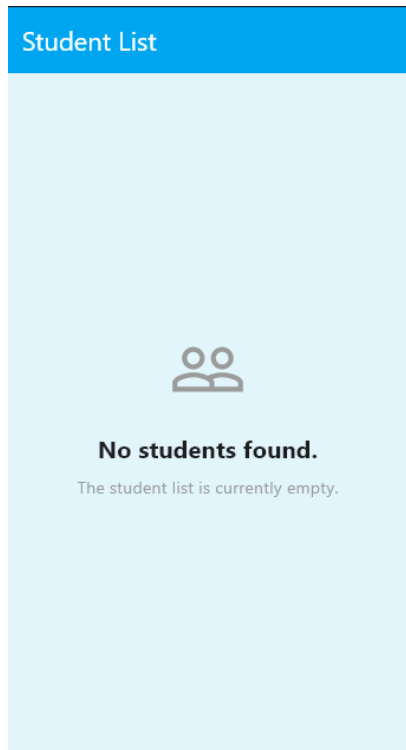
Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list
- Empty-state UI
- Application still functioning correctly

```
const List<Student> students = [];
```



```
body: students.isEmpty
  ? const Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Icon(
          Icons.people_outline,
          size: 65,
          color: Colors.grey,
        ), // Icon

        SizedBox(height: 20),

        Text(
          'No students found.',
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
            color: Colors.black87,
          ), // TextStyle
        ), // Text
      ],
    ),
  )
```

🚩 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

- Aaron

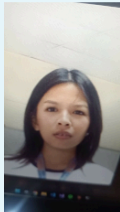
- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order

```
final List<Student> sortedStudents = List<Student>.from(students)
    ..sort((a, b) => a.name.compareTo(b.name));
```

Student List	Student List	Student List
 Chim Boop BSIT 2nd Year Age: 20 Hobby: Hobby Tanan ID: 202300126 chim.boop@school.edu Favorite Subject: Networking	 Julia Yashika Kong BSIT 3rd Year Age: 20 Hobby: Depende sa uso ID: 202300128 julia.kong@school.edu Favorite Subject: GEC	 Karylle Buta BSIS 3rd Year Age: 21 Hobby: Magpakatanga ID: 202300127 karylle.but@school.edu Favorite Subject: Information Systems
 Mary Jocelyn Syllanto BSIT 3rd Year Age: 20 Hobby: Drinking Secret ID: 202300123 mary.syllanto@school.edu Favorite Subject: Math	 Maryan Gridwan BSIT 3rd Year Age: 21 Hobby: Playing Mobile Legends ID: 202300124 maryan.gridwan@school.edu Favorite Subject: Networking	 Mav Sumalangitka BSCS 1st Year Age: 19 Hobby: Dancing ID: 202300125 mav.sumalangitka@school.edu Favorite Subject: Recess

🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
PS C:\Users\Admin\source\repos\IT314AB_SYLLANTO4\IT314AB_SYLLANTO\my_first_flutter_project_emji> git branch
* feature/student-list
  main
PS C:\Users\Admin\source\repos\IT314AB_SYLLANTO4\IT314AB_SYLLANTO\my_first_flutter_project_emji> git add .
warning: in the working copy of 'my_first_flutter_project_emji/linux/flutter/generated_plugin_registrant.cc',
LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_project_emji/linux/flutter/generated_plugin_registrant.h',
LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_project_emji/linux/flutter/generated_plugins.cmake', LF will
be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_project_emji/macos/Flutter/GeneratedPluginRegistrant.swift',
LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_project_emji/windows/flutter/generated_plugin_registrant.c
c', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_project_emji/windows/flutter/generated_plugin_registrant.h',
LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'my_first_flutter_project_emji/windows/flutter/generated_plugins.cmake', LF
will be replaced by CRLF the next time Git touches it
PS C:\Users\Admin\source\repos\IT314AB_SYLLANTO4\IT314AB_SYLLANTO\my_first_flutter_project_emji> git commit
-m "feat: submit activity 4:feature/student-list"
[feature/student-list cdff39b] feat: submit activity 4:feature/student-list
7 files changed, 195 insertions(+), 109 deletions(-)
create mode 100644 my_first_flutter_project_emji/assets/chim.PNG
create mode 100644 my_first_flutter_project_emji/assets/jolya.JPG
create mode 100644 my_first_flutter_project_emji/assets/krylle.PNG
create mode 100644 my_first_flutter_project_emji/assets/marianne.JPG
create mode 100644 my_first_flutter_project_emji/assets/mav1.JPG
PS C:\Users\Admin\source\repos\IT314AB_SYLLANTO4\IT314AB_SYLLANTO\my_first_flutter_project_emji> git push origin feature/student-list
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 8 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 713.22 KiB | 33.96 MiB/s, done.
Total 12 (delta 3), reused 4 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
```